KASRA SIAVASHPOUR

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EDUCATION

Sharif University of Technology (SUT), Tehran, Iran 2020–Jul 2026
Bachelor of Science, Computer Science, Department of Mathematical Sciences
Allameh Tabatabaei High School (selective, non-governmental), Tehran, Iran 2016–2020
High School Diploma, Mathematics and Physics

RESEARCH INTERESTS

- Distributed & Decentralized Systems
- IoT & Cyber-Physical Systems
- Multi-Agent Systems
- AI-Driven Network Automation

MANUSCRIPTS

Submitted

- A. Vaezi, S. Niknejad, **K. Siavashpour**, A. Dadras, "FAIR: A Novel Envy-Free Multi-Agent Model for Traffic Systems," submitted to the **International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2027**.

In Preparation

- A. Siavashpour, **K. Siavashpour**, M. Azadegan, B. Safaei, H. Peyvandi, M. Imran, "A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems," to be submitted to the **IEEE Open Journal of the Communications Society (OJ-COMS)**.
- A. Vaezi, A. Daneshgar, A. Dadras, **K. Siavashpour** (author order not final), "The Loop of the Rings: A Fully Decentralized Reliable Cooperative System," manuscript in preparation for submission to the **IEEE Symposium on Foundations of Computer Science (FOCS) 2027**. [arXiv preprint]

RESEARCH EXPERIENCE

Research Assistant, **Institute for Research in Fundamental Sciences (IPM)**
& **Sharif University of Technology**, Tehran, Iran May 2026–present
Supervisor: Dr. Arash Vaezi
Project: "FAIR: A Novel Envy-Free Multi-Agent Model for Traffic Systems"

- Developed an event-driven traffic simulator and evaluated FAIR on road networks from Berlin, London, and Shanghai (up to 15K edges and 93K vehicles) against a Google Maps-style navigation baseline.
- Proposed road-occupancy and pre-entry travel-time rules adopted in the final model; also identified a dimensional inconsistency in an earlier cost equation.
- Extended the trajectory-level envy-freeness theorem with per-vehicle bounds and waiting/detour terms; identified counterexamples to two preliminary lemmas, leading to their reformulation.
- Under heavy congestion, FAIR cleared the network 2.5–10× faster than the baseline and improved travel-time predictability across all three cities.

Research Assistant, **Institute for Research in Fundamental Sciences (IPM)**
& **Sharif University of Technology**, Tehran, Iran Aug 2024–present
Supervisors: Dr. Arash Vaezi, Dr. Amir Daneshgar
Project: "The Loop of the Rings: A Fully Decentralized Reliable Cooperative System"

- Analyzed LoR's randomized ring structure to evaluate protocol reliability under adversarial trader behavior.
- Investigated reliability and user satisfaction in networks with at least 60% honest traders.
- Designed and conducted 421 simulations with 500 traders each to evaluate protocol robustness and validate theoretical findings.
- Implemented the core protocol in Go and evaluated its behavior under worst-case adversarial conditions.

Research Assistant, **Sharif University of Technology**, Tehran, Iran Aug 2024–present
Supervisors: Dr. Hossein Peyvandi, Dr. Masoumeh Azadegan (Khatam University)
Collaborators: Dr. Bardia Safaei, Dr. Muhammad Imran (University of Glasgow)
Project: "A Comprehensive Survey of Self-Organizing Networks in Next-Generation Cellular Systems"

- Co-wrote the survey with A. Siavashpour in equal shares, covering 114 references; the other authors advised and reviewed.
- Classified 106 papers by SON function and period: self-optimization accounts for 61%, and self-healing grew from a single paper (2014–16) to 24.
- Identified open problems, including conflicts between SON functions and SON's role in 6G.

Research Intern, **Politecnico di Milano**, Milan, Italy (Remote) May 2024–Aug 2024
Supervisor: Dr. Gianni Antichi

- Researched WebAssembly for systems acceleration and designed a benchmarking plan for CPU, memory, latency and throughput.

Research Intern, **Sharif University of Technology**, Tehran, Iran Oct 2023–Jan 2024
Supervisor: Dr. Bardia Safaei

- Simulated a decentralized flying ad hoc network (FANET) of UAVs in NS-3 with ant-colony-based cluster routing.
- Implemented beaconing and neighbor discovery under UAV mobility, and analyzed energy–delivery trade-offs.
- Built reproducible pipelines (configs, seeds, scripts) for experiment reruns.

NOTABLE PROJECTS

SliceDB: Distributed Key-Value Store *(Go, etcd)* [GitHub Link] 2025, extended 2026
Course project with A. Siavashpour

- Hash-partitioned key-value store with write-ahead-log replication from leaders to followers, etcd leases for membership and a fenced controller election.
- Online resharding under load (8 to 16 partitions); leader handoff loses no acknowledged write.
- `kill -9` of a node under load: 0 client-visible errors (16 retries), writes paused about 4 s.

Multimodal Product Search *(Python, CLIP, FAISS)* [GitHub Link] 2025–2026
Course project, Machine Learning Systems Design

- Text and image retrieval over Torob's catalog with multilingual sentence embeddings, CLIP and FAISS (text index of 1.02M products, image index of 100K), tuned with Optuna and evaluated on 5,000 logged user queries.
- Added a chat agent that uses retrieval as a tool and keeps per-conversation memory.

Highly Available HDFS + Spark *(Hadoop, ZooKeeper, Spark)* [GitHub Link] 2025
Course project with A. Siavashpour

- Deployed HDFS with automatic NameNode failover and ran Spark queries on NYC taxi data.

Go Judge: Online Judge *(Go, PostgreSQL, Docker)* [GitHub Link] 2025
Course project with A. Siavashpour

- Built the web server, UI and database schema; each submission runs in a Docker sandbox.

TEACHING EXPERIENCE

Teaching Assistant, Sharif University of Technology

- **Computer Networks** — Dr. Mahdi Dolati — Fall 2026, Winter 2026
- **Operating Systems** — Dr. Hossein Asadi — Fall 2026
- **Operating Systems** — Dr. Hossein Asadi & Dr. Rasul Jalili — Fall 2024
- **Computer Networks** — Dr. Hossein Peyvandi — Winter 2024
- **Operating Systems** — Dr. Arash Vaezi [Piazza] — Winter 2024
- **Computer Networks** — Dr. Kambiz Mizanian — Fall 2023
- **Operating Systems** — Dr. Mohaddeseh Mirbeygi & Dr. MohammadAli Mirzaei — Fall 2023

Designed and graded weekly assignments (120+ students in the largest classes), wrote rubrics, held office hours, guided projects and built autograders.

HONORS & AWARDS

Top 0.1% among 155,250 candidates, Iranian University Entrance Exam (B.Sc.) — 2020
1st place in National Hovercraft Competition (high school) — design & control — 2016–2017

SKILLS

Programming Languages: Go, Python, Java, C/C++, Bash

Libraries: PyTorch, scikit-learn, FAISS, NumPy, Matplotlib

Systems & Tools: etcd, Docker, PostgreSQL, Spark, NS-3, Git

Languages: Persian (Native), English (C1)